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Supplementary Materials for

Leaping out of the water: aerial-aquatic locomotion with flapping wings

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Materials and Methods

Robot structure

For fast experimental aerial-aquatic experiments, modularity is key to study different physical parameters and to repair the robots. The entire structure can be disassembled down to the motor and reduction system, thanks to the ubiquitous use of M2 screws. The wings can be removed thanks to two bolts at the root level. The square tail tube is held through compression and its length can be adjusted, or exchanged entirely. Lastly, the robots are optionally fitted with force sensor mounts or sliding mounts for fixed or linearly constrained experiments. It is critical to minimize mass in flight. The wing flapping engine is the heaviest component at 55 g, this includes the motor, reduction and crankshaft. The total added mass for water adaptation is 13 g, representing a mass cost of 5.2% compared to a flight-only vehicle, lower than existing aerial-aquatic design *(37)*. A detailed component list is available Table S1 and mass breakdown in Fig. S1.E. In the presented design, the only cavity is that of the USB-C port required for programming. The 1.5 mm-thick carbon fuselage of the robot (Fig. 1.E) features cutouts, permitting fast water emptying upon take-off while also reducing structural weight. The double plate design offers high stiffness, supporting the internal crankshaft loads.

Wings and coatings

In flapping flight, wing mass should be minimized, first to lower inertial losses, and second to provide adequate passive pitch phasing *(38)*. To this end, the wings consists of a carbon skeleton and a 3-mm diameter leading edge spar, covered with a nylon fabric (Fig. S2.C). Adhesive and a single sewing point at extremities maintains the membrane attached to the skeleton in high-load aquatic conditions. The wing skeleton consists of four 0.8 x 3 mm bars. The inner diagonal bar links the leading edge to the tail, where is attached via a flexible hinge. It resists in-plane torques on the shoulder through buckling when frontal impact on the wing occur, eliminating the risk of damage to the driving mechanism. The outer diagonal bar provides chord stiffness, controlling the resulting deformation of the membrane (Fig. S3.A, Fig. S3.B, Fig. S3.C). The two radial bars are membrane extenders, giving 180% surface increase over a triangular wing. This wing structure offers rapid manufacturing (<30 min of assembly time), enabling quick design iterations. In contrast to birds, it forgoes camber at the benefit of symmetry. This design trait is chosen to provide symmetric forces (active downstroke and upstroke) in underwater flight.

The transition from water to air is challenging in part due to the added mass from the wet surfaces. By far the largest surface on the flyers is the wing, and hydrophobicity is observed in bird wings. Early 20th century hypotheses proposed that uropygial gland secretions function to impart a hydrophobic coating to feathers *(39)*. This idea is supported by observations that aquatic bird species tend to possess relatively larger uropygial glands *(40,41)*, suggesting a potential role of preen oil in enhancing water repellency *(42)*. However, subsequent research indicates that preen oil is not essential for water repellency per se, but rather serves as a feather conditioner, maintaining feather integrity, strength, and flexibility until the next moult *(43)*. Feather microstructure—particularly the barbules—plays a critical role in water repellency *(44)*, allowing aquatic birds to shed water quickly. We test superhydrophobic treatments to reduce the added mass on synthetic wings. While the nylon fabric covering the wing is light (8% body weight), it wets

easily, increasing the wing mass by almost 100%. We experimentally measure the effect of nano-particle surface coatings (2-step Ultra Everdry) for this hybrid application (Fig. S2.B). Three configurations were compared: uncoated, spray-painted light coating and brush-painted heavy coating. We observe that with superhydrophobic coatings, the mass of water carried by the wing upon vertically lifting it out of the water is reduced to less than 25% of the dry wing mass. However, the coating itself adds unwanted mass. A light coating is a good trade-off between low additional coating weight when dry (+ 13%) and substantial weight saving when wet (-30%), and is used in experimental results shown here.

Wing sets

We tested three sets of wings of different absolute sizes while maintaining a constant aspect ratio. Experiments were conducted both in air and in water. To facilitate manufacturability and structural consistency, all wings share an identical internal skeleton: the same rods are used across sizes. As a result, smaller wings contain proportionally more structural material relative to their area, whereas larger wings contain less. Consequently, the effective stiffness decreases with increasing wing size (Fig S2. A).

To enable comparison across wing sizes, we use spanwise effective stiffness Π_1 *(45, 46)*, defined as the ratio of elastic bending resistance to fluid dynamic loading:

$$\Pi_1 = \frac{EI}{\rho U^2 c^3},$$

where E is Young's modulus of the wing material, I is the second moment of area of the wing spar, ρ is the fluid density, U is the characteristic translational velocity, and c is the wing chord length.

For comparison with avian wings, we use scaling laws for bird wing bones *(47)*. Humerus length l_{humerus}, diameter d_{humerus} and wall thickness t_{humerus} are inferred from wingspan s , assuming isometric geometric scaling.

$$l_{\text{humerus}} = 0.11s$$
$$d_{\text{humerus}} = 0.32 l_{\text{humerus}}^{0.8}$$
$$t_{\text{humerus}} = 0.1 d_{\text{humerus}}$$

The resulting values were computed for flight in air and for swimming in water, considering both extended and folded wing configurations.

To independently vary effective stiffness while holding wing geometry constant, we fabricated five additional sets of wings of identical (medium) size but with different wing spar diameters (2.0, 2.5, 3.0, 3.5, and 4.5 mm). Varying spar diameter modifies the second moment of area I without altering wing area or aspect ratio, allowing systematic exploration of stiffness effects.

In addition to spanwise effective stiffness, we also investigate spanwise torsion (i.e., chordwise bending), which plays a critical role in flapping-wing propulsion. Spanwise torsion was quantified experimentally by tracking the relative motion of the leading and trailing edge during flapping in both air and water (Fig. S3.A). Assuming a flat plate between the two edges, this relative displacement was used to compute an equivalent local twist angle as a function of time over the

flapping cycle (Fig S3.B,C). In air, the leading–trailing edge exhibits an approximately 60 degree phase offset, whereas in water the phase offset is approximately 180 degree. These measurements were obtained at low forward velocities, and we expect the phase shift to decrease at higher forward speeds during cruising flight or swimming.

Flapping actuation

The flapping of the wings require a powerful actuation method which can propel the body regardless of the medium density, and tolerating immersion. While the two opposing operating points: high-velocity low-torque in air and low-velocity high-torque in water could be accommodated with a gearbox, which has been proposed in *(48)*, this would be prohibitively heavy given the mechanical power transmitted here. Instead, we choose to implement a brushless motor drive with feedback which enables us to study the entire frequency spectrum. We design the fuselage to minimize the body cross-section, to 24 x 48 mm by aligning the motor and reduction behind the right wing root.

At scales above 10 g, electromagnetic actuation provides the highest energy density while satisfying torque and speeds constraints. For aerial-aquatic operation, brushless motors are the most adequate method as they inherently do not have exposed contacts and can therefore be submersed. Testing here occurs in fresh water and adaptation to sea water is left for future work.

Flapping wings follow a reciprocating motion, whereas motor actuation is rotating. Two methods have been proposed for this transformation. Direct-drive methods rely on reversing motor direction periodically, supplemented by a parallel spring to recover part of the wing energy. Crank-shaft methods on the other hand produce an approximate sine-wave output from a constant motor speed but feature a more complex mechanical coupling (Fig. S1.A). The latter is selected due to our findings showing that this configuration better minimizes frontal area, and therefore body drag in water.
Researchers have suggested increased efficiency at a single resonance frequency *(49)*, however it is unclear if animals leverage resonance *(50)*. Since operating at resonance can only occur at a single frequency, while hybrid locomotion dictates two different mode, elastic energy storage is discarded from this design. The motor is selected to have the largest diameter (minimum kV) below 20 g (10% robot weight), limiting the reduction required *(51)*. The AT-2303 motor's shaft is removed and replaced with a new shaft with a press-fit 3-mm pinion gear that fits in the 23:1 planetary gearhead. The gearhead is connected to a crankshaft-linkage system (or four-bar linkage) which both converts the rotational motion into reciprocal motion and offers a further reduction of 2:1. The linkage pivots around an offset connection point to the left wing root gear. The motion of the left wing is symmetrically transmitted to the right wing through meshed gears. As this entire transmission system is intended for full immersion, careful practical considerations on material choices are further detailed in Fig S1.C.

The resulting wing amplitude is 60 degrees, a choice consistent with Atlantic puffin's aerial amplitude *(3),* and also constrained by the dimensions of the fuselage. An average dihedral angle of 4° offers passive roll stability and is a generally accepted value in fixed-wing engineering.

A closed-form solution of the forward kinematics $\varphi_{\text{wing}} = f(\varphi_{\text{motor}})$ of the transmission mechanism (Fig. S1.A and Fig. S1.B) for the wing flapping motion can be derived analytically (see Fig. S1.A for variable definitions):

$$f(\varphi_{motor}) = 180° - \left(\arccos\left(\frac{w_f(\alpha)}{sgn\left(v_f(\alpha)\right)\sqrt{u_f(\alpha)^2 + v_f(\alpha)^2}} \right) + \arctan\left(\frac{u_f(\alpha)}{v_f(\alpha)} \right) \right) - \epsilon\,,$$

with

$$\alpha = \frac{\varphi_{\text{motor}}}{r}$$

$$u_f(\alpha) = -2ac\sin(\alpha)$$

$$w_f(\alpha) = b^2 - a^2 - c^2 - x_3^2 + 2ax_3\cos(\alpha)$$

While the mass proportion of pectoral to supracoracoideus muscles is greater in diving birds by a factor of 2.1 or more *(52)*, our design provides symmetric motor efficiency and power. This is unnecessary in flight but results in an active upstroke in water. This has also been shown to be used by alcids *(53)*.

Flapping with Field-Oriented Control

Brushless motors excel at power density (over 5kW/kg) yet require individual control of three electrical phases. If the torques are low, the rotor can be assumed to remain synchronized with the rotating magnetic field, a strategy employed in most aerial flyers nowadays. Here, the water environment exerts loads beyond this threshold, and de-synchronization occurs. To overcome this limitation, we use Field-Oriented Control (FOC) *(54)*. It requires feedback from a coated hall encoder sensor shown in Fig. S1.D. The high-level control architecture is shown in Fig. S1.F. While this solves high-torque aquatic conditions, in aerial conditions rotational speeds are high and bottlenecks occur at the sensor reading speed level and computing capability, an issue worsened by the high amount of pole pairs in the motor. This technical challenge is solved through hardware (10 MHz SPI sensor) and software enhancements (dynamic electric angle), see High speed FOC. Our driver can provide 0 to 15200 rpm speed at the motor level, i.e. up to 11 Hz flapping frequencies.

Electronics

To comply with the low volume buoyancy constraint, individual electronic components need to tolerate immersion and should therefore be sealed individually (see method in Fig. S2.D). The custom flight controller of assembly is cast in a flexible silicoset compound, resulting in a 7g encapsulated sub-system. Schematic and manufacturing details available in Fig. S1.C. It sends commands to and receives data from the motor electronic speed controller (ESC), which is similarly sealed. Onboard, we collect power and flapping data locally at 50 Hz sample frequency, which is transferred wirelessly after the submerged experiments.

Swimming performance

The robots were allowed to navigate freely within a shallow water pool measuring 8×4×0.4m. The velocity and position of the robot is tracked optically with a magnetically ceiling-mounted GoPro 12 (120 fps, linear) and a CSRT (OpenCV) tracking tool. The image is captured with a rectilinear perspective correction. This ensures that length is maintained across the frame. Refraction error is minimized by having the robot swim as close to the surface as possible (20 cm depth), and by having the camera located directly above the robot. We repeated every swimming experiment five times to capture experimental uncertainty, and every video was manually trimmed to ensure the robots were swimming straight and unconstrained when tracked. Velocity data was derived by differentiating the recorded positional data. We present the maximum cycle-average speed here, with standard deviation obtained from 5 separate swims. The power at the battery level is measured onboard the robot thanks to a dedicated current sensor (ACS 723). The current sensor is connected to an analog input on the microcontroller and sampled at 100 Hz. The data is stored in memory and transmitted to the user after a flight/swim via Bluetooth BLE. The cost of transport is calculated as follows:

$$COT = \frac{P}{mgV}$$

where P is the input power, m the mass of the system and V the velocity.

Transition experiments

We followed a staged approach towards successful transition: first rail-supported transitions (Movie 1) and secondly unaided transitions. The rail consisted of a custom slider system which constrained the direction of motion to an inclined line (Fig S4.B), thus eliminating control requirements and providing partial support in the air to study the vehicle's performance (velocity, frequency, electrical power). The 3-m long rail is half submerged at a 30 degree angle. The flapping is controlled in frequency (motor rotational velocity) with a torque limit. We evaluate the capability of the robot to move from water to air and how the water sheds from the wings. For unaided rail-free transition, the robot was inserted below the water surface on a short 0.2 m support. The robot is free to slide out of the support's end under its own propulsion. This support is mounted on a heavy tripod, at a fixed angle to the horizontal between 45° and 90°. The robot-holding tripod rested on the lake bed, next to the camera-holding tripod that shows the state of the wings as it transitions out of the water. The transition maneuver was triggered by a timer, which activated the motor and initiated wing flapping at 10 Hz for a duration of 5 seconds (Fig S4.A4).

Aerial experiments

The power of the robots is electrically measured onboard, and is streamed over the robot's wireless link. The force and torque values are given by an ATI Nano25 six-axis force/torque sensor, interfaced with a National Instruments NI-DAQmx 9.5.1 data acquisition system (sampling rate: 1000 Hz; data averaging: 5 samples). This setup was mounted on a Stäubli TX-90 robotic arm (maximum angular positioning error: <0.1°) and positioned in front of an open-jet wind tunnel at 20 degrees angle of attack (test volume: 1.92 × 1.68 × 1.5 m^3; turbulence intensity: <1%, WindShape, Switzerland). The wind tunnel was operated at the robot's expected mean flight velocity of 6 m/s. The wind speed was measured with a hot wire anemometer to ensure consistent

speeds across the wingspan (within 0.1 m/s). This is in line with the free fight average tracked velocity of 6.3 m/s (Fig. S5.D), where flight speed was computed from position recorded using an Optitrack motion capture system at 180 frames per second. For the free flights, the robot was hand-launched.Flight range is calculated based on power measurements in our wind tunnel: $P_{flight,\ 6Hz,\ 6\ m/s} = 18.54W$. This gives us a runtime of the battery at this power is:

$$t_{max} = \frac{350\ [mAh] * 3.6 * 14.8\ [V]}{18.54\ [W]} = 1005\ s \sim 17\ min$$

And a distance of $D_{fly} = Vt = 6.0$ km.
In swimming, at a power level of $P_{swim,\ 1Hz,\ 0.49\ m/s} = 4.53$ W, this would be reduced to: $D_{swim} = 2.0$ km.

Diving

The robot is accelerated to a flight speed of 5 m/s using an electronically-controlled launch system mounted 0.5 m above the water surface. The robots are held by the launcher at 20 degrees angle of attack via a top-mounted sliding adapter. The launch system is suspended from the ceiling so as to not interfere with the tail (Movie S3). The launch sequence is manually triggered and the robot's flapping starts after the robot has left the rail. Synchronization is performed manually. The robot then covers a distance of 3-4 m in the air before entering the water in the middle of the tank. Gliding phases are possible thanks to the ability to lock the wing at an angle with the position controller. Underwater flapping starts on a timer, which could be replaced by an input from the onboard sensor, and stops after 5 seconds underwater. The velocity is tracked optically with a magnetically ceiling-mounted GoPro 12 (120 fps, wide angle) and our OpenCV tracking tool.

Flapping

Flapping propulsion is ubiquitous in all actively flying and fast swimming animals. In synthetic systems, it provides a number of benefits over traditional propeller-powered aircrafts. In aerial applications, flapping-wing systems provide enhanced safety due to the absence of exposed rotating blades, which vastly reduces the risk of injury during human interaction or collision *(55)*. The inherently low tip-speed of flapping wings also leads to quieter operation, which is desirable in both civilian and wildlife monitoring contexts. Furthermore, flapping propulsion allows for high maneuverability and broad flight envelopes *(56)*, including the ability to generate lift at low speeds during take-off or hovering and to transition to efficient forward flight at higher speeds *(57)*. The latter is particularly important in the context of take-off from the water (low speed) to forward flight (high speed). In aquatic environments, flapping propulsion confers robustness to mechanical impacts and entanglement *(58)*. Flexible fins are more tolerant to collisions with debris or obstacles compared to rigid propellers, which are prone to damage. This is a key benefit in cluttered, shallow, or structurally complex environments such as coral reefs *(59),* riverbeds, or kelp forests.

High speed FOC

Regular drones ESCs (Hobbyking 12A and EP Aer11-Aer-Series 11A have been tested) start losing synchronization at low speed. This is a problem for water propulsion. Field-oriented control (FOC) ESCs tend to be more complex and larger. The B-G431B-ESC1 development board is a good compromise and brings programming flexibility. Rated at 30A. High rotational speed is not possible by default, prohibiting its use for aerial flapping flight. Three issues for increasing FOC velocity exist. The first concerns the communication speed with the hall-effect encoder. The second is the speed of the micro-controller loop. The third one is due to delays from the rotation sensor.

The default connection of hall effect sensor is over I2C, which limits the speed at which the motor can be driven. To go beyond 600 rad/s, we switch to an SPI communication. This is not supported by the G431-ESC1 board by default. The difficulty is that there is no SPI link exposed directly on the board. The solution is to use the SPI3 bus, which is present on the board, and accordingly updated software. The SPI bus speed should also be increased to 10 MHz. The AS5047P sensor is indicated for speeds at up to 28 kRPM and selected for this application. It is pin and code-compatible with the more common AS5048 sensor.

By switching to the SPI3 bus, we lose the ability to reprogram. The ST-Link (break-away part of the board) needs to be connected to the microcontroller for programming. That is the case when out of the box via Serial2, but not anymore when those GPIOs are used as SPI3 for the sensor. In that case, we need to connect Serial1 to the ST-Link. This can be achieved by disconnecting USART2 and connecting USART1, removing R23 and R24 (on the daughter board, desolder). Then wires should be soldered to the now free pins of R23/24 (the pads closer to U3, further away from the USB port). Finally, connect the wire from R23 to the pad for hall sensor 2 (PB7). And connect the wire from R24 to the pad for hall sensor 1 (PB6) on J8. The pins need to be remapped for SPI usage (can be done by overwriting WEAK PinMap in the file pinmap_custom.h). We seek to maximize the main loop frequency, which through the adaptations in this document can be increased to 13 kHz. Changing the motor.motion_downsample = 3, improves loop frequency by 0.8 kHz. Disabling monitoring increases loop frequency by another 0.2 kHz. And switching to SVPWM should theoretically reduce computing load but has been shown to reduce loop speed in practice. Lastly, tuning the inductance value precisely helps L=0.0001. Compensating for the sensor lag as speed increases is necessary to go beyond 600 rad/s without current losses. This value is hand-tuned every 100 rad/s at no load, and a fitted function is used for operation on the robot.

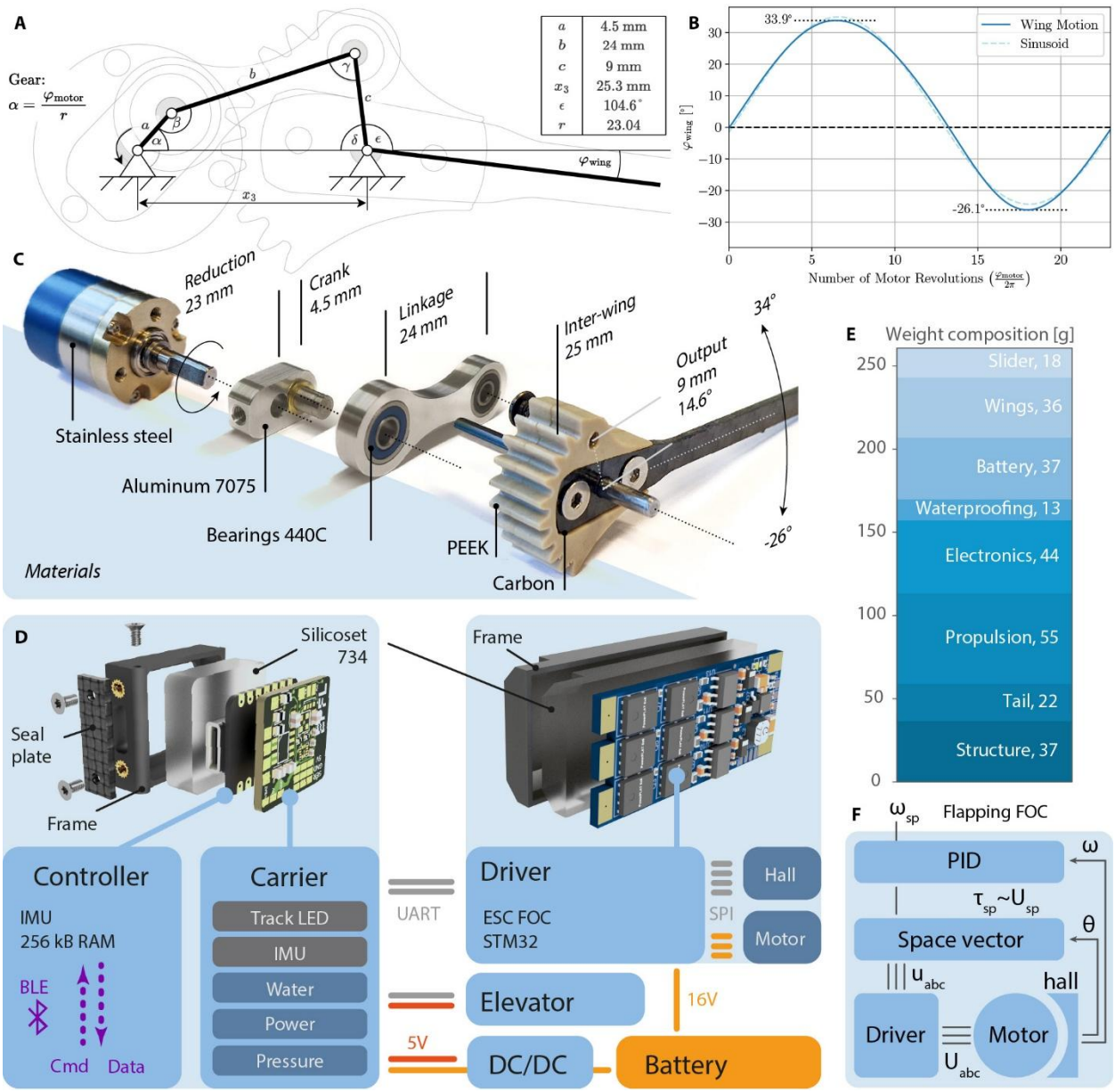


Fig. S1. Flapping actuation. A. The motor (not shown) drives a 23 mm double-stage planetary reduction (23:1). Its 4 mm output D-shaft, supported by a bearing, drives a 4.5 mm crank via an M3 insert screw. The crank press-fits a 4 mm shaft rotating in the 440C bearing of the 24 mm linkage. Both parts are CNC-cut from 7075 aluminum. The linkage carries two sealed bearings and drives the left wing root. Wing roots use a router-cut PEEK gear and a 1.5 mm carbon-fiber plate, joined by two countersunk M3 screws. The custom gear mounts the 9 mm linkage shaft and connects to the right wing (not shown). **B** Electronics**.** A custom PCB with an nRF52840 (Bluetooth 5.0) measures IMU data, water presence, current, voltage, and depth (unused). It connects to the motor driver, elevator servo, and battery/regulator. The brushless driver communicates over UART and stores velocity data. An AS5047P encoder provides feedback over a 10 MHz SPI link. **C**. Mass. Waterproofing adds 13 g. The 37 g battery accounts for only 15% of the total mass. **D**. Flapping control. Field-oriented control enables low-speed flapping,

while high-speed sensors and software allow operation up to 1600 rad/s. Wing motion follows the setpoint ω_sp. Space-vector decomposition generates the three-phase drive signal u_abc, which is applied by the MOSFET driver as U_abc at 13 kHz.

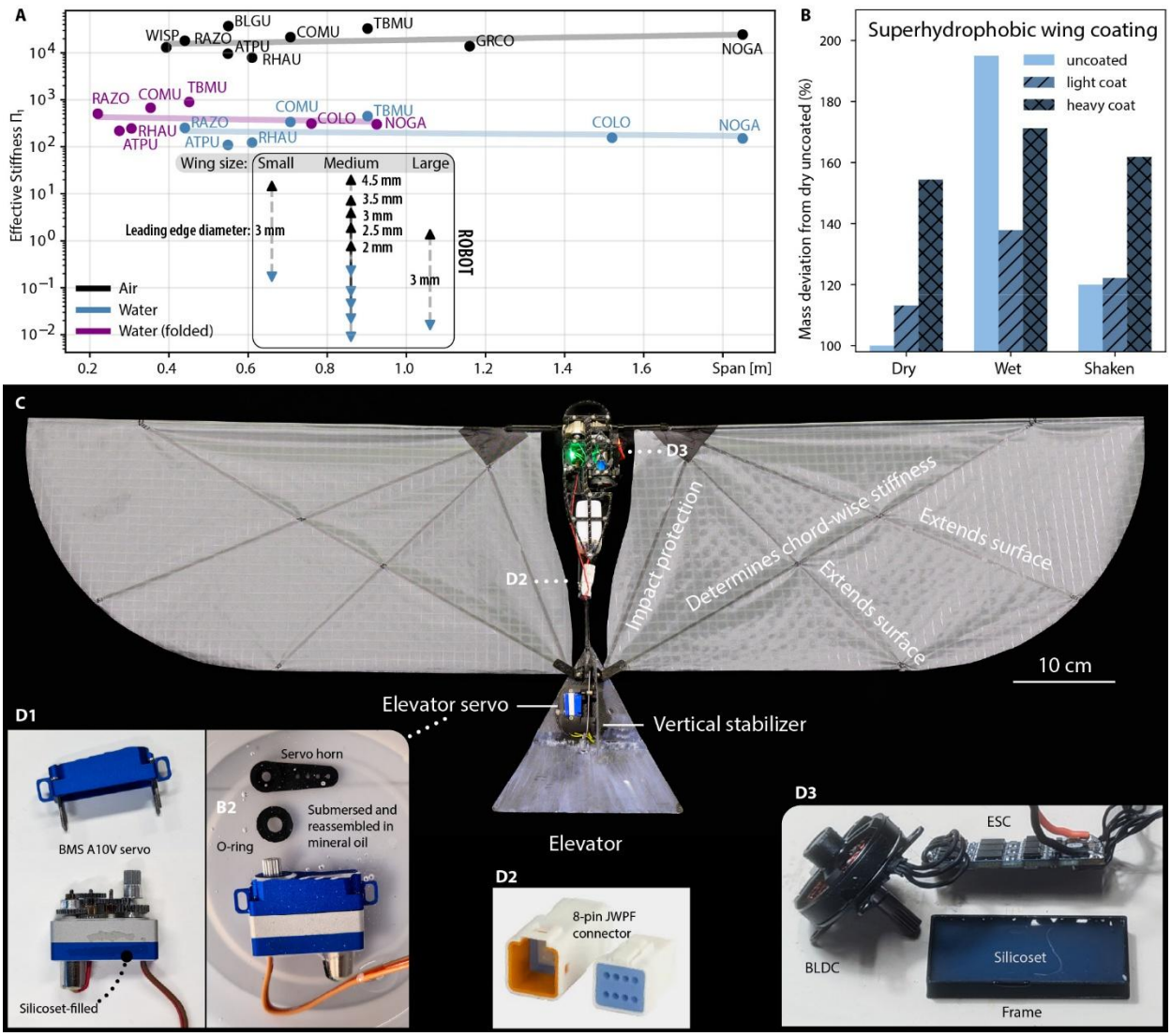


Fig. S2. Wing and robot design. A. To enable comparison across bird species and with the robotic platform, we evaluate effective stiffness. Values are shown for flight in air (black) and swimming in water with wings extended (blue) and folded (purple). Relative to birds, the robot exhibits greater wing flexibility; this increased compliance allows operation at higher flapping frequencies in water, enabling the air–water transition. **B.** Influence of superhydrophobics coatings on the mass of the wings. This illustration shows how much weight coatings add to the mass of the wings. This is shown for both a light and a heavy coating. The wings are then measured in 3 conditions: dry, wet and shaken. A wet wing is pulled out of the water. A shaken wing is manually given two hard shakes equivalent to a flapping motion. The mass is given as a proportion of the dry, uncoated wings. We observe that a lightly-coated wing is the best option for a wing that occasionally gets wet. **C.** Top-down view of the open-wing designs. **D.** The tail actuator is a BMS-A10V servo, waterproofed with a hobby-grade method, adequate for freshwater shallow testing. First, the servo's cable side is opened, filled with silicoset compound, and cured after closing. Then the gear side cover is removed. The whole servo assembly is immersed in mineral oil, with an o-ring seal added between servo and horn. While immersed, the horn is mounted and the servo re-assembled. The ESC and autopilot boards are potted in

silicoset, poured into a 3D-printed frame before insertion. The battery connects to the PCB via an 8-pin JWPF connector. The batteries are potted in silicoset.

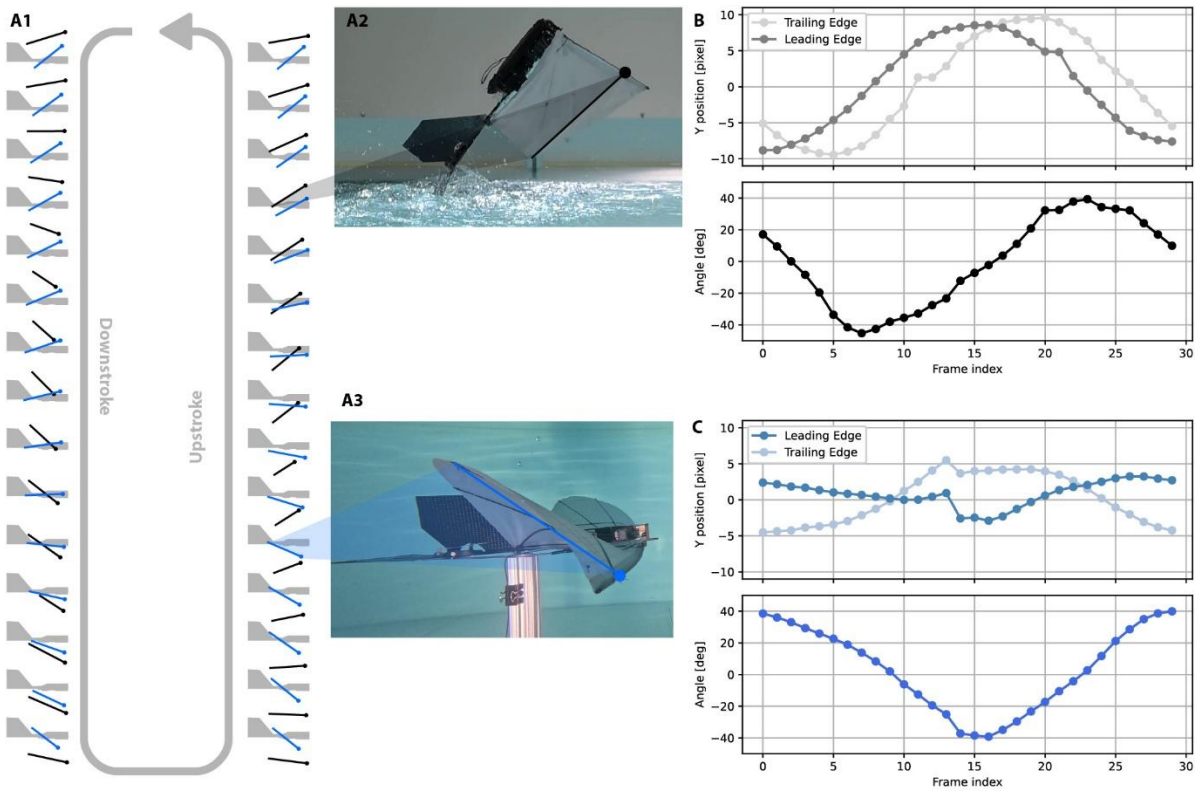


Fig. S3. Wing pitch deformation. Wing deformation is crucial in flapping wing flight. **A.** One Flapping cycle illustrating chordwise wing deformation in air (black) and water (blue). **B.** Time histories of leading- and trailing-edge motion through vision-based tracking during flight in air at 10 Hz, showing an approximately 60° phase offset. **C.** Corresponding leading- and trailing-edge measured in water at 1 Hz, exhibiting an approximately 180° phase offset.

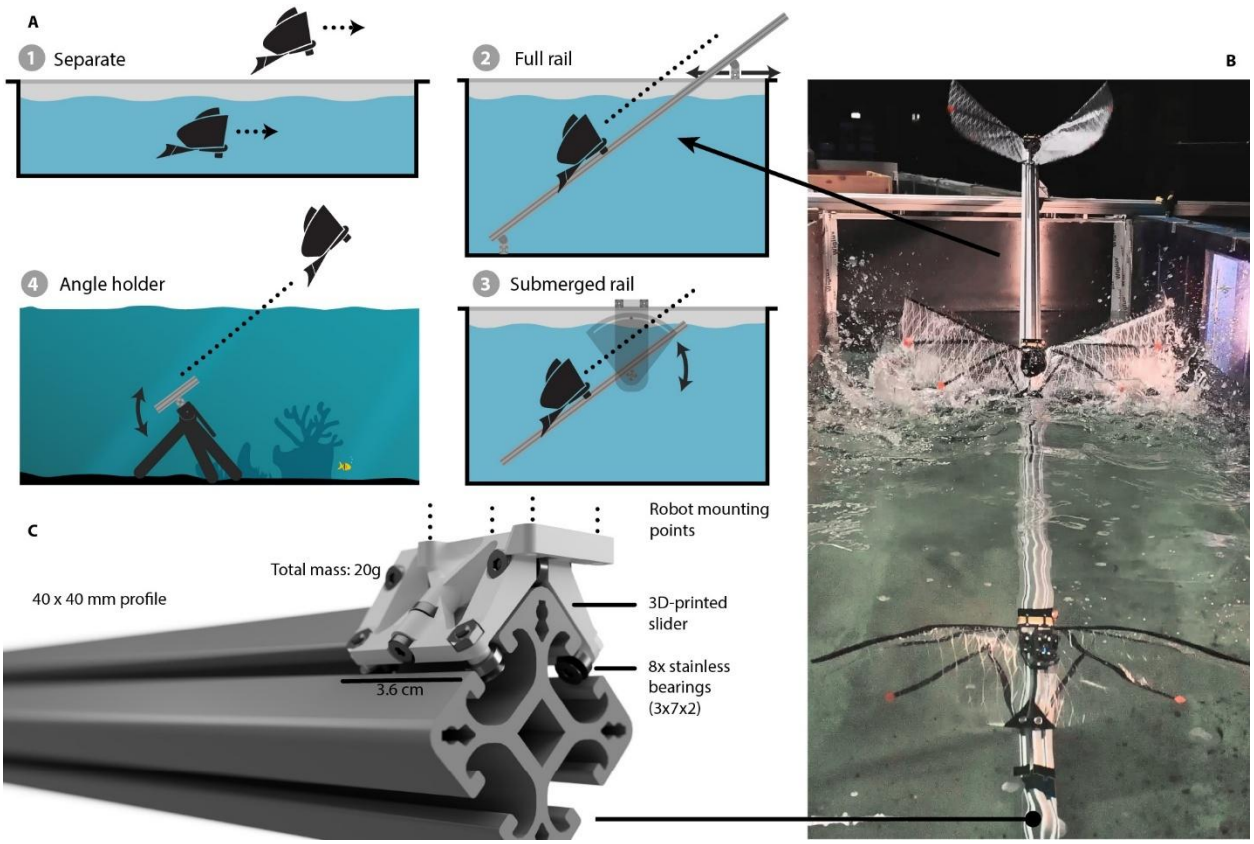


Fig. S4. Development process using a rail. A. Development process followed to obtain the transition results in this work. 1 The robots were operated in separate media, allowing investigation of wing sizes, efficiency, and development of most engineering work required for flight (high frequency) and swimming (high torque, waterproofing). 2. A deeper tank accommodates our 3 m inclined rail from water to air. The robots slide on this rail, with room for 1 m swimming and 1 m aerial distance. The angle is adjusted by moving the top transversal beam mounted on the tank edge, indicated by the double arrow. 3. A smaller rail is mounted on a submerged beam. The angle and depth can be freely adjusted. The distance from the rail top to the surface is effectively a free-swimming distance. 4. The robots are initially held underwater on a short rail fixed to a large tripod to maintain a known launch angle. **B.** Composite image showing the robots in 3 different positions on the rail: underwater (bottom), upon transition (middle) and in the air (top). The robots are self-propelled solely through the flapping of their wings. Note: the wing membrane is different to that of the results shown elsewhere in this work, which is one of the elements that we iterated over in this experiment. **C.** Rendered view of the custom-developed 18g slider system for our aerial aquatic robots to be tested on a rail. The rail is a standard 40 x 40 mm profile. The polycarbonate slider relies on 8x stainless steel 7 mm bearings to roll on the edge of the rail. This is required to resist the large transversal forces from flapping. The edge is chosen to ensure the flapping wings cannot collide with the rail. While commercials polymer-based sliders exist with much lower weight, they cannot provide motion when perpendicular forces are applied.

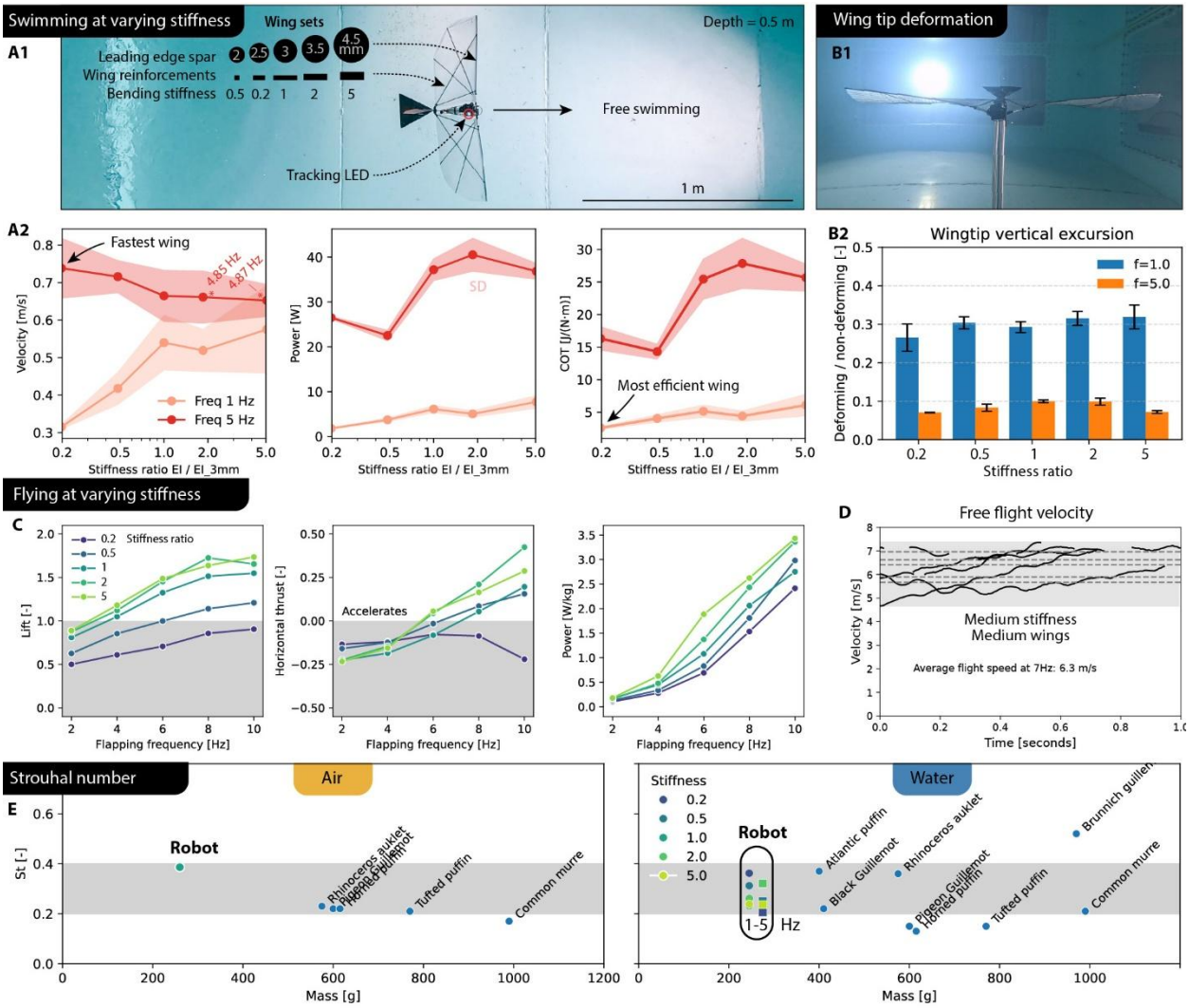


Fig. S5. Impact of flexibility. **A.** Swimming speeds, power and cost of transport of the robot with varying wing stiffness. Speed and power mean and std deviation is calculated from 4 experiments. **B.** Fixed deformations experiment. (B2) average wingtip vertical excursion (full stroke). At 5 Hz, the deformation is such that the wingtips travel only roughly 10% of the non-deforming stroke. **C.** Results from wind tunnel measurements with flexible wings (5 different stiffness ratios: 0.2,0.5,1,2,5). **D.** Flight velocities of multiple flights shown with their average values when flapping at 7 Hz in air. The minimum flight speed is 4.6 m/s and the maximum is 7.4 m/s. **E.** Strouhal number based on flight cruising conditions in air (left) and tracked forward speed and wingtip excursion in water (right). The gray band represents the 0.2-0.4 value which has been found in most swimming, flying and undulating animals, and correlates with high propulsive efficiency *(60)*.

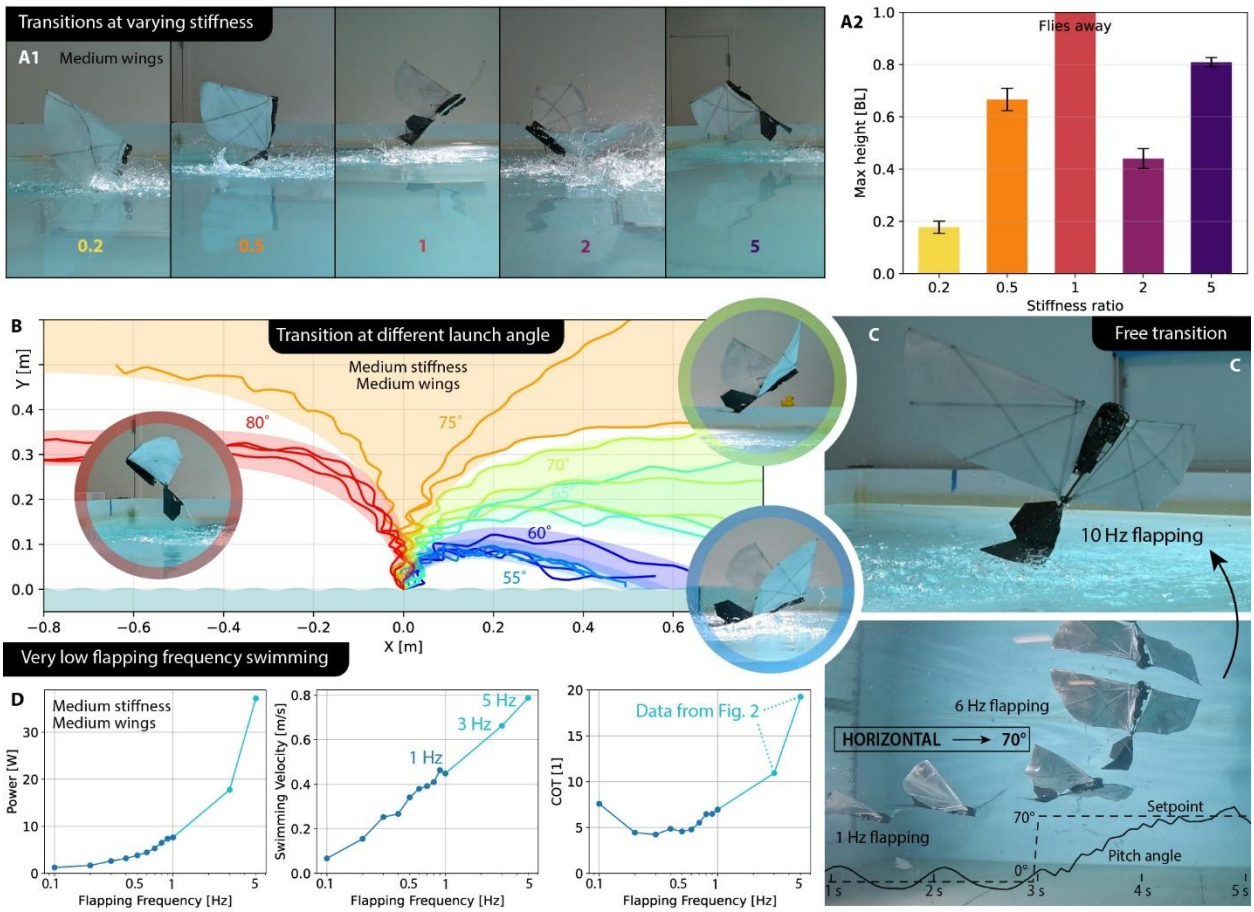


Fig. S6. Extended transitions and low frequency swimming. A. Effect of wing stiffness on inclined-support based take-off performance: (A1) representative snapshots of the transition for different stiffness ratios, and (A2) corresponding maximum take-off height, revealing an optimal intermediate wing compliance. **B.** Trajectories of the robot during the air--water transition for different support launch angles using medium-stiffness wings, highlighting the coupled role of initial pitch and passive wing deformation. **C.** Example of a free transition from water to air without any inclined support, driven solely by flapping at 10 Hz. **D.** Swimming performance at very low flapping frequencies for medium-stiffness wings, including power consumption, swimming velocity, and cost of transport (COT), demonstrating efficient aquatic propulsion enabled by passive wing deformation.

Table S1. List of components.

Category	Component	Material/Model	Nb	Mass	Total
Structure	Frame	Carbon 1.5mm plate (swiss-composite)	2	4.7	9.4
	Vertical panels	Onyx with cont. fiber (Markforged)	2	6.2	12.4
	Motor holder	Onyx with cont. fiber (Markforged)	1	4.3	4.3
	Battery holder	Onyx with cont. fiber (Markforged)	1	1.1	1.1
	Tail holder	Onyx with cont. fiber (Markforged)	1	4.3	4.3
	Screws	M2 Stainless	40	0.13	5.2
Tail	Tail	Carbon 0.5 mm plate (Swiss-composite)	1	22.2	22.2
Propulsion	Crank	Al 7075	1	1.3	1.3
	Linkage	Al 7075, with bearings	1	4.3	4.3
	Reduction	Series 20/1R 1:23 (Faulhaber)	1	38	38
	Right plate	Carbon 1.5mm plate (Swiss-composite)	1	1.5	1.5
	Left wing gear	PEEK 5 mm (RS-Pro)	1	2.8	2.8
	Right wing gear	PEEK 5 mm (RS-Pro)	1	6.7	6.7
Electronics	Flight controller	Seeeduino NRF52840	1	2.4	2.4
	Carrier board	Custom PCB	1	1	1
	Servo	BMS A10V (Blue Bird Model)	1	9.2	9.2
	Step-down reg.	5V/3A (generic)	1	0.8	0.8
	ESC	G431-ESC1 (STMicroelectronics)	1	5.2	5.2
	Motor	AT2303 1500kV (T-Motors)	1	17	17
	Hall sensor	AS5047P (ams-OSRAM) @10MHz SPI	1	1.2	1.2
	Cables		1	7	7
Waterproof.	Seeeduino	Silicone + 3D print	1	3.7	3.7
	ESC	Silicone + 3D print	1	5.2	5.2
	Connector	JWPF 08 (JST)	1	3.9	3.9
Battery	Battery	4S 350 mAh LiPo (Swaytronics)	1	37	37
Wings	Wings	Carbon structure (Swisscomposites)	2	14	26
	Membrane	Ripstop Nylon 20D, 40 g/m2 (Extremtextil)	2	3	6
	Coating	Ultra Ever Dry 2-component	2	2.1	4.2
	Bonding	3M VHB tape + stitches at the ends			
Slider	Carriage	PC 3D print	1	7.5	7.5
	Screws / bearing	Stainless	1	10	10

Table S2. Bird acronyms.

Acronym	Common Name	Acronym	Common Name
SGPE	South Georgia Diving Petrel	**COEI**	Common Eider
COPE	Common Diving Petrel	**WCPE**	White-chinned Petrel
MAMU	Marbled Murrelet	**LESC**	Lesser Scaup
DOVE	Dovekie	**ANHI**	Anhinga
ANMU	Ancient Murrelet	**LIPE**	Little Penguin
ATPU	Atlantic Puffin	**SGSH**	South Georgia Shag
HOPU	Horned Puffin	**EUSH**	European Shag
RHAU	Rhinoceros Auklet	**DCCO**	Double-crested Cormorant
PIGU	Pigeon Guillemot	**SOGP**	South Giant Petrel
RAZO	Razorbill	**NOGA**	Northern Gannet
EUCO	Eurasian Coot	**CSPE**	Chinstrap Penguin
TUPU	Tufted Puffin	**COLO**	Common Loon
GCGR	Great Crested Grebe	**MAPE**	Macaroni Penguin
COMU	Common Murre	**ADPE**	Adélie Penguin
BRGU	Brünnich's Guillemot	**TBMU**	Thick-billed Murre
COME	Common Merganser	**LAGU**	Laughing Gull
		PECO	Pelagic Cormorant

Table S3. Robot parameters.

Parameter	Robot	Alcid (ATPU)	Unit
Tip-to-tip span	0.89	0.55	m
Chord	0.20	0.07	m
Area	0.17	0.04	m^2
Weight	0.26	0.40	kg
Wing loading	1.5	10.8	kg/m^2
Aspect ratio	4.4	8.8	
Dihedral	3.5	NA	deg
Aerial amplitude	60	90	deg
Flapping frequency	0.1–11	3.1–9.2	Hz
Tail distance	0.2	NA	m

Table S4. Robot comparison.

ID	Robot	Mass [kg]	COT	Medium	Reference
a	Temple LEAP swimmer	0.15	0.78	water	*61*
b	Multi-functional Microrobot	0.04	0.2	water	*62*
c	Yale ART	0.32	0.94	water	*63*
d	iSplash	0.32	11.53	water	*64*
e	Tunabot Flex	0.26	4.65	water	*65*
f	Tunabot Flex	0.26	3.96	water	*66*
g	Tuna-inspired Robot	0.52	1.51	water	*67*
h	Wire-driven active fish	0.31	1.35	water	*68*
i	Group of robotic fish	0.45	1.28	water	
j	Robotics water beetle	0.1	1.2	water	*69*
k	Tensegrity robotic fish	0.36	0.87	water	*70*
l	Cownose-ray-like fish	0.25	0.66	water	*71*
m	Median-finned robot	0.58	0.61	water	*72*
n	IPMC robotics fish	0.27	0.45	water	*73*
o	Self-powered soft robot	0.22	0.34	water	*74*
p	MFC PZT fish	0.24	0.31	water	*75*
q	Robotic zebrafish	0.15	0.26	water	*76*
r	Finned unicore	0.09	3.66	water	
	This work: water	0.4	1.98	water	
	This work: air	0.4	15.78	air	
s	E-Flap	0.72	8.33	air	*77*
t	Delfly	0.2	35.0	air	*78*
u	Robo Raven IV	0.63	15.87	air	*79*
v	Dove	0.5	16.0	air	*80*
w	X-wing ornithopter	0.2	30.0	air	*81*

Partially collected with (82)

Movie S1. Rail-assisted water exit (indoor experiment).

The flapping-wing aerial–aquatic robot performs a transition from underwater propulsion to aerial flapping flight while exiting the water surface in an indoor water tank. The robot is guided along a 3-m-long inclined rail (30°) made from a standard 40 × 40 mm aluminum profile. A custom 18 g polycarbonate slider equipped with eight 7-mm stainless steel bearings rolls along the rail, constraining the direction of motion and partially supporting the robot's weight during the maneuver. The video shows the transition in real time (left) and slowed down 5× (right).

Movie S2. Tank take-off (indoor experiment).

The robot exits the water surface in an indoor tank under controlled conditions. The vehicle is initially guided toward the surface using a rigid support inclined at 65° relative to the horizontal, after which it is released and continues the transition under its own propulsion. The video is slowed down 5× relative to real time.

Movie S3. Plunge-diving (indoor experiment).

The robot performs a plunge-diving maneuver in an indoor water pool (8 × 4 × 0.4 m). The sequence shows the acceleration phase using a launcher to 5 m/s suspended 1 m above the pool, followed by a short aerial phase (both gliding and flapping aerial phases are tested), water entry, and subsequent underwater swimming driven by the flapping wings.

Movie S4. Water exit of the flapping-wing aerial–aquatic robot in a natural lake (1).

The robot transitions from underwater propulsion to aerial flapping flight while exiting the water surface at a 65° angle relative to the horizontal. The angle is set by an underwater rail that is mounted on a tripod. The heavy-duty tripod rests on the lake floor. The maneuver demonstrates the capability of the flapping-wing system to generate sufficient thrust for a successful water–air transition in an outdoor environment. The video is slowed down 10× relative to real time.

Movie S5. Water exit of the flapping-wing aerial–aquatic robot in a natural lake (2).

The robot transitions from underwater propulsion to aerial flapping flight while exiting the water surface at a 55° angle relative to the horizontal. The maneuver demonstrates the capability of the flapping-wing system to generate sufficient thrust for a successful water–air transition in an outdoor environment. The video is slowed down 10× relative to real time.

Movie S6. Water exit of the flapping-wing aerial–aquatic robot in a natural lake with pitch controller (3).

The flapping-wing aerial–aquatic robot exits the water surface in a natural lake (Lausanne, Switzerland) at a 65° angle relative to the water plane while using an active pitch controller to stabilize the transition from underwater propulsion to aerial flapping flight. The maneuver is first shown in real time from a distant viewpoint, followed by a close-up view slowed down 10×.